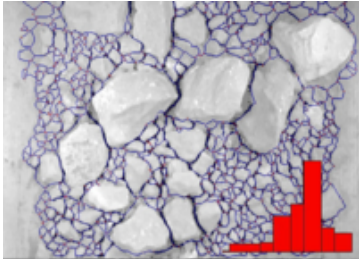


Mesh (scale)

Mesh is a measurement of particle size often used in determining the particle-size distribution of a granular material. For example, a sample from a truckload of peanuts may be placed atop a mesh with 5 mm openings. When the mesh is shaken, small broken pieces and dust pass through the mesh while whole peanuts are retained on the mesh. A commercial peanut buyer might use a test like this to determine if a batch of peanuts has too many broken pieces. This type of test is common in some industries, and, to facilitate uniform testing methods, several standardized mesh series have been established.

Metal surfaces mechanically polished are designated as having a mechanical finish related to the abrasive used.

Granulometry

Basic concepts
Particle size · Grain size Size distribution · Morphology
Methods and techniques
Mesh scale · Optical granulometry Sieve analysis · Soil gradation
Related concepts
Granulation · Granular material Mineral dust · Pattern recognition Dynamic light scattering

Many mesh sizes were historically given in the number of holes per inch; due to the width of the wires in the mesh, mesh numbers did not correspond directly to fractional inch sizes, and several different systems standardized with slightly different mesh sizes for the same mesh numbers.

Commercial dimensions

Commercial sieve mesh dimensions

Sieve size	Opening		Standard Mesh		Tensile Bolting Cloth			Mill Grade			Market Grade		
(mm)	(in)	(µm)	US	Tyler	Mesh	Opening	Wire	Mesh	Opening	Wire	Mesh	Opening	Wire
5.60	0.22	5600	3.5	-	-	-	-	-	-	-	-	-	-
4.75	0.187	4750	4	-	-	-	-	-	-	-	-	-	-
4.00	0.157	4000	5	-	-	-	-	-	-	-	-	-	-
3.35	0.132	3350	6	-	-	-	-	-	-	-	-	-	-
2.80	0.110	2800	7	-	-	-	-	-	-	-	-	-	-
2.36	0.093	2360	8	-	-	-	-	-	-	-	-	-	-
2.00	0.079	2000	10	-	-	-	-	-	-	-	-	-	-
1.70	0.0661	1700	12	10	14	0.062	0.009	12	0.065	0.018	12	0.0603	0.023
1.40	0.0555	1400	14	12	16	0.0535	0.009	14	0.054	0.017	14	0.051	0.0204
1.18	0.0469	1180	16	14	18	0.0466	0.009	16	0.0465	0.016	16	0.0445	0.0181
1.00	0.0394	1000	18	16	22	0.0380	0.0075	18	0.0406	0.015	18	0.0386	0.0173
0.85	0.0331	850	20	20	24	0.0342	0.0075	20	0.0360	0.014	20	0.034	0.0162
0.71	0.0278	710	25	24	28	0.0282	0.0075	24	0.0287	0.013	24	0.0277	0.014
0.60	0.0232	600	30	28	34	0.0229	0.0065	30	0.0238	0.0095	-	-	-
0.50	0.0197	500	35	32	38	0.0198	0.0065	34	0.0204	0.009	30	0.0203	0.0128
0.47	-	470		-	40	0.0185	0.0065	36	0.0188	0.009	-	-	-
0.465	-	465		-	42	0.0183	0.0055	38	0.0178	0.0085	-	-	-
0.437	-	437		-	44	0.0172	0.0055	-	-	-	35	0.0176	0.0118
0.425	0.0165	425	40	35	46	0.0162	0.0055	40	0.0165	0.0085	-	-	-
0.389	-	389		-	48	0.0153	0.0055	-	-	-	40	0.0150	0.0104
0.368	-	368		-	50	0.0145	0.0055	-	-	-	-	-	-
0.355	0.0139	355	45	42	52	0.0137	0.0055	45	0.0142	0.008	-	-	-
0.310	-	310		-	60	0.0122	0.0045	50	0.0125	0.0075	-	-	-
0.300	0.0117	300	50	48	62	0.0116	0.0045	55	0.0112	0.007	-	-	-
0.282	-	282		-	64	0.0111	0.0045	-	-	-	50	0.0110	0.0090
0.270	-	270		-	70	0.0106	0.0037	-	-	-	-	-	-
0.260	-	260		-	72	0.0102	0.0037	-	-	-	-	-	-
0.250	0.0098	250	60	60	74	0.0098	0.0037	60	0.0102	0.0065	-	-	-
0.241	-	241		-	76	0.0095	0.0037	-	-	-	-	-	-
0.231	-	-	-	-	78	0.0091	0.0037	-	-	-	60	0.0092	0.0075
0.212	0.0083	212	70	65	84	0.0084	0.0035	-	-	-	-	-	-
0.193	-	-	-	-	90	0.0076	0.0035	-	-	-	-	-	-
0.180	0.0070	180	80	80	94	0.0071	0.0035	-	-	-	80	0.0070	0.0055
0.165	-	-	-	-	105	0.0065	0.0030	-	-	-	-	-	-

0.150	0.0059	150	100	100	120	0.0058	0.0025	-	-	-	100	0.0055	0.0045
0.125	0.0049	125	120	115	145	0.0047	0.0022	-	-	-	120	0.0046	0.0037
0.105	0.0041	105	140	150	165	0.0042	0.0019	-	-	-	150	0.0041	0.0026
0.090	0.0035	90	170	170	200	0.0034	0.0016	-	-	-	180	0.0033	0.0023
0.075	0.0029	75	200	200	230	0.0029	0.0014	-	-	-	200	0.0029	0.0021
0.063	0.0024	63	230	250	-	-	-	-	-	-	250	0.0024	0.0016
0.053	0.0021	53	270	270	300	0.0021	0.0012	-	-	-	270	0.0021	0.0016
0.044	0.0017	44	325	325	-	-	-	-	-	-	325	0.0017	0.0014
0.037	0.0015	37	400	400	-	-	-	-	-	-	400	0.0015	0.0010
0.025	0.0010	25	500	-	-	-	-	-	-	-	500	0.0010	0.0010
0.020	0.0008	20	625	-	-	-	-	-	-	-	625	0.0008	-
0.015	0.0006	15	800	-	-	-	-	-	-	-	800	0.0006	-
0.010	0.0004	10	1250	-	-	-	-	-	-	-	1250	0.0004	-
0.005	0.0002	5	2500	-	-	-	-	-	-	-	2500	0.0002	-

Equivalent mesh sizes from 5 µm to 25.4 mm also exist.^{[1][2][3]}

Available sieve sizes are usually regulated by standards. Those in common use are ISO 565:1990 and ISO 3310-1:2000 (international),^[4] EN 933-1(European)^[5] and ASTM E11:01 (US). EN standards are available with national 'badging' so appear as BS EN, FR EN, DE, etc.

Practical sizes

Although such information contains long lists of sieve sizes, in practice sieves are normally used in series in which each member sieve is selected to pass particles approximately $1/\sqrt{2}$ smaller in diameter or 1/2 smaller in cross-sectional area than the previous sieve. For example the series 80mm, 63, 40, 31.5, 20, 16, 14, 10, 8, 6.3, 4, 2.8, 2 mm is routinely available in many European countries or the series with the larger steps 63, 31.5, 16, 8, 4, 2, 1 mm, 500 µm, 250, 125, 63 µm is commonly used to grade aggregates in the construction industry. Such series are somewhat derived from the principles originally established by Renard and now known as Renard series. Some users replace some of those indicated above with 45, 22.4, 12.5, 11.2 and 5.6 mm sieves, mostly because of historical usage of such sizes in their country or industry.

References

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3. "Particle Size - US Sieve Series and Tyler Mesh Size Equivalents" (<https://www.azom.com/article.aspx?ArticleID=1417>). *www.azom.com*. Retrieved 2020-10-13.
4. ISO 3310-1:2000 Test sieves — Technical requirements and testing — Part 1: Test sieves of metal wire cloth (<https://www.iso.org/standard/25147.html>) *www.iso.org*, accessed 13 October 2020
5. DIN EN 933-1 Tests for geometrical properties of aggregates - Part 1: Determination of particle size distribution - Sieving method' (<https://www.en-standard.eu/din-en-933-1-tests-for-geometrical-properties-of-aggregates-part-1-determination-of-particle-size-distribution-sieving-method/>) *www.en-standard.eu*, accessed 13 October 2020

External links

- A comparison of widely used sieve size specifications (<https://web.archive.org/web/20120314084653/http://www.nexlab.ru/files/equipment/spec.pdf>)

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